Unit 4 Guided Notes — Geometry: Angles & Triangles

Name _____ **Date** _____ **Period** _____

Name the angle pair. Write an equation. Solve for x.

🎯 **By the end of this unit, I can…**

- Use angle relationships formed by parallel lines and a transversal
- Apply the Triangle Angle-Sum and Exterior-Angle Theorems
- Solve for unknown angles by writing and solving equations

Standard: NC.8.G.5

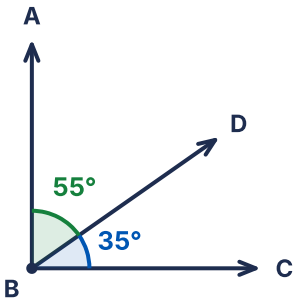
Fill in the blanks as we go. For every example and problem, **write each step on its own line** in the space given, and box your final answer.

1 Angle Relationships

Relationship	Definition	Sum / Rule
Complementary	Two angles that add to _____ °	_____
Supplementary	Two angles that add to _____ °	_____
Vertical Angles	_____	_____
Linear Pair	_____	_____

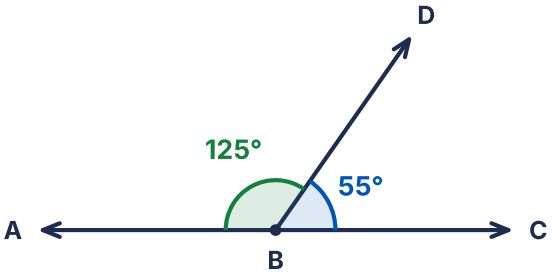
Each angle is named with three letters — the vertex is the MIDDLE letter (e.g. ∠ABD has vertex B). Arcs show which angle each measure belongs to.

COMPLEMENTARY = 90°



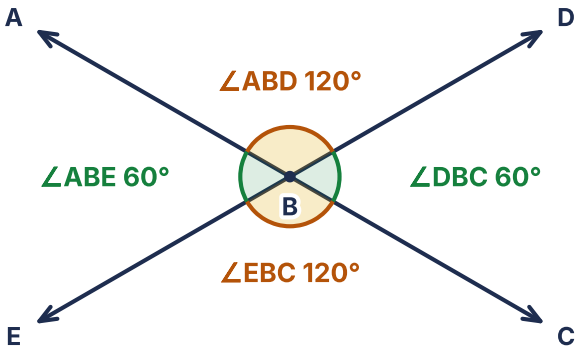
35° + 55° = 90°

SUPPLEMENTARY = 180°



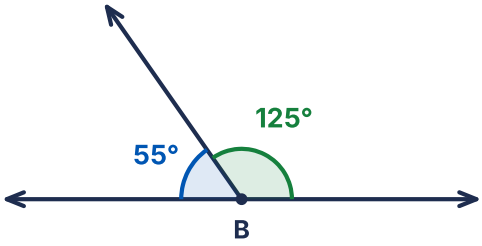
55° + 125° = 180°

VERTICAL ANGLES



Opposite angles are EQUAL: ∠ABD = ∠EBC, ∠DBC = ∠ABE

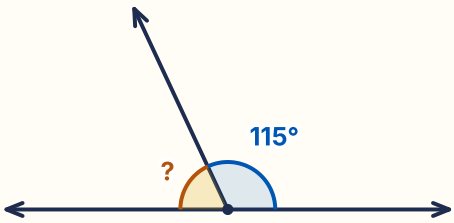
LINEAR PAIR = 180°



side by side on a line: 125° + 55° = 180°

Quick Check

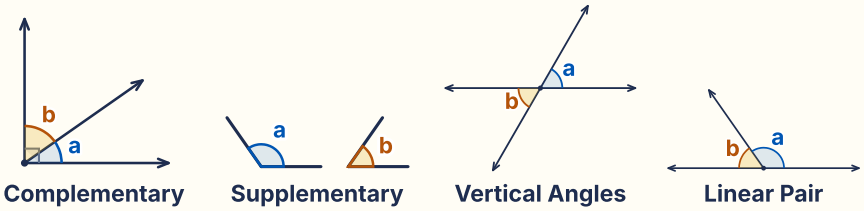
MC Two angles form a linear pair. One measures 115°. What is the other?



- ○ 65°
- ○ 75°
- ○ 245°

Match the Relationship to its Rule

MATCH Write the letter of the matching rule on each line.



Complementary | Supplementary | Vertical Angles | Linear Pair

_____ 1. Complementary

_____ 2. Supplementary

_____ 3. Vertical Angles

_____ 4. Linear Pair

A. Always EQUAL

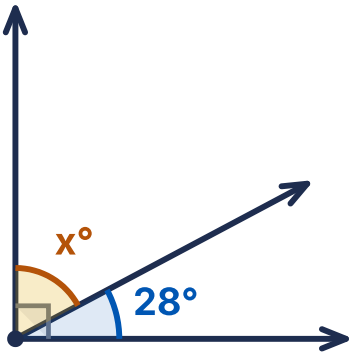
B. Add to 90°

C. Adjacent angles on a line → 180°

D. Add to 180°

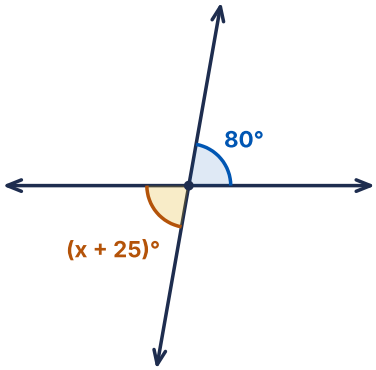
✏ Practice: Angle Relationships

1. ★ Try it The two angles are complementary. Find x.



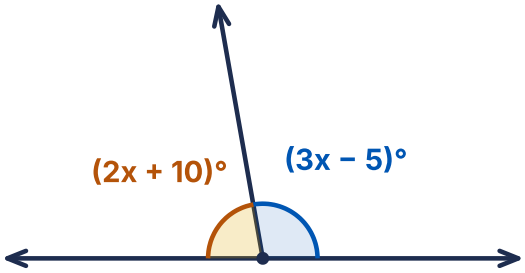
Answer: _____

2. ★★ On level Two lines cross. Find x.



Answer: _____

3. ★★★ Challenge The two angles form a linear pair. Find x.



Answer: _____

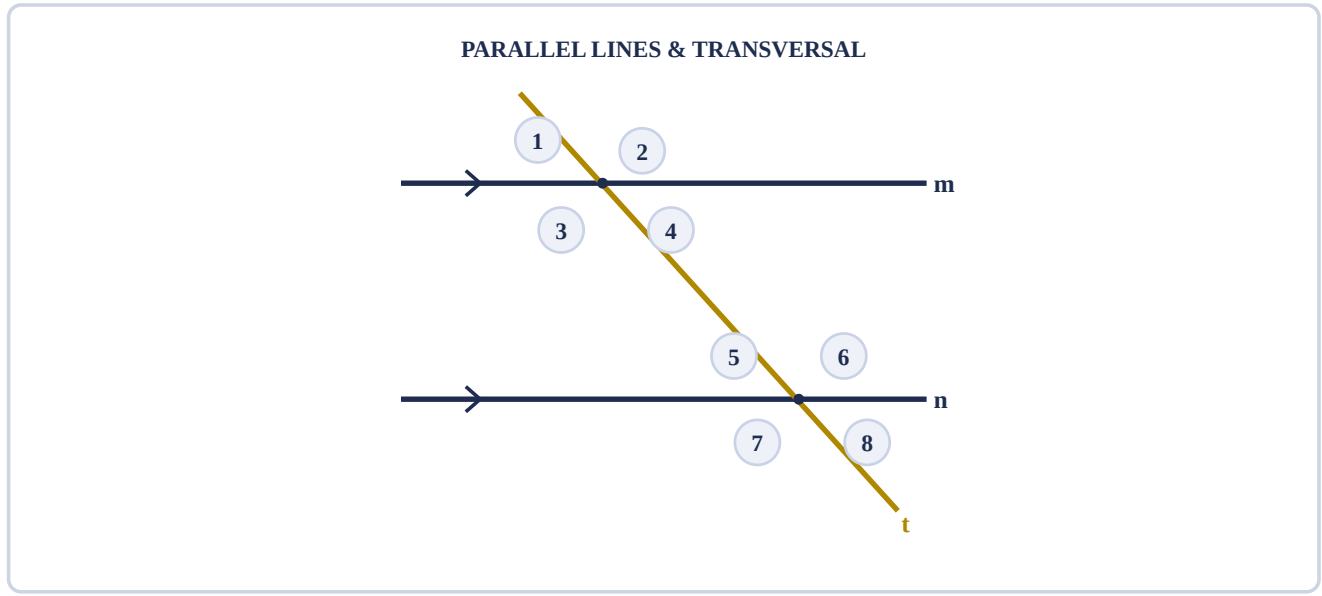
2 Parallel Lines & Transversals

📖 **Words to know**

Parallel lines: lines that never meet.

Transversal: a line that crosses two or more lines.

📌 **REMEMBER:** Corresponding, Alternate Interior, Alternate Exterior, and Vertical angles are CONGRUENT (equal). Same-Side Interior and Same-Side Exterior angles are SUPPLEMENTARY (add to 180°).

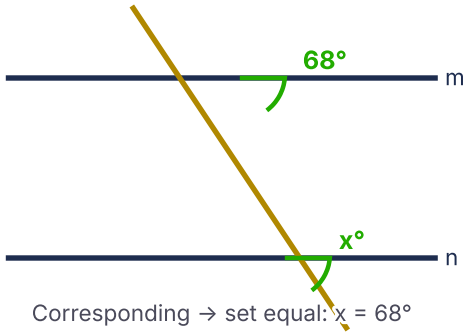


Two parallel lines (m ∥ n) cut by transversal t form 8 angles.

- Corresponding: (1, 5), (2, 6), (3, 7), (4, 8)
- Alternate Interior: (3, 6), (4, 5)
- Alternate Exterior: (1, 8), (2, 7)
- Same-Side Interior: (3, 5), (4, 6)

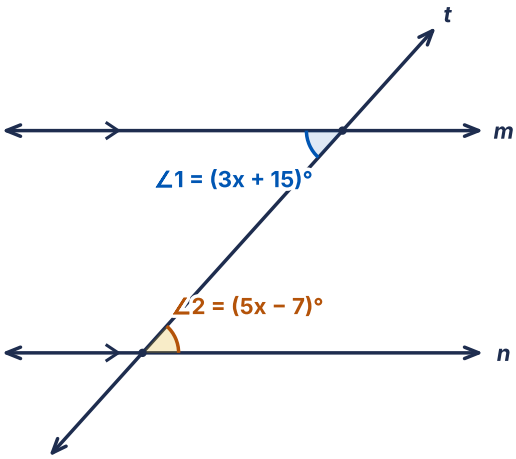
Angle Pair Type	Relationship	Rule to Use	Example
Corresponding	_____	_____	68° → x = 68°
Alternate Interior	_____	_____	101° → x = 101°
Alternate Exterior	_____	_____	142° → x = 142°
Vertical	_____	_____	120° → x = 120°
Same-Side Interior	_____	_____	134° → x = 46°
Same-Side Exterior	_____	_____	77° → x = 103°

CORRESPONDING ANGLES ARE EQUAL



Corresponding → set equal: x = 68°

Worked Example — ∠1 = 3x + 15 and ∠2 = 5x − 7 (alternate interior angles)



∠1 and ∠2: between the lines, opposite sides of t

Alternate interior angles are EQUAL:

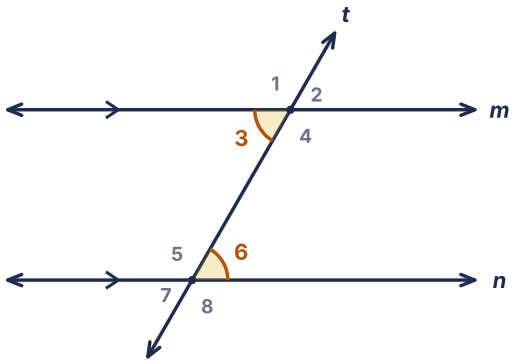
Subtract 3x:

Add 7:

✓ Divide by 2:

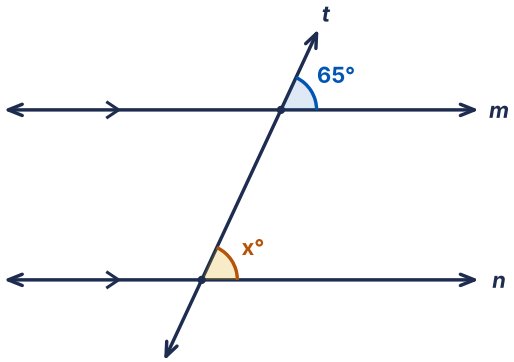
✏ Practice: Parallel Lines & Transversals

1. ★ Try it Lines m and n are parallel. What kind of angle pair are ∠3 and ∠6?



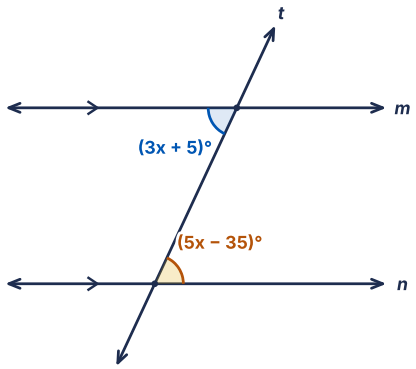
- ○ Corresponding
- ○ Alternate interior
- ○ Same-side interior

2. ★ Try it Lines m and n are parallel. Find x.



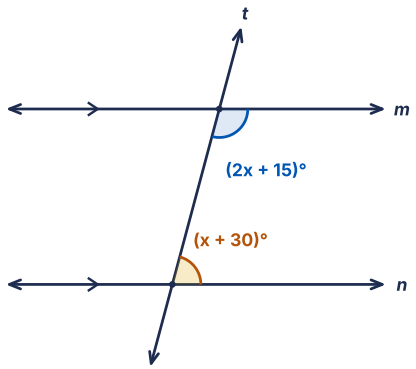
Answer: _____

3. ★★ On level Lines m and n are parallel. Find x.



Answer: _____

4. ★★★ Challenge Lines m and n are parallel. Find x. Then find the measure of the angle marked (2x + 15)°.

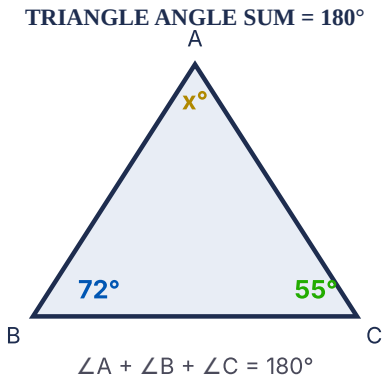


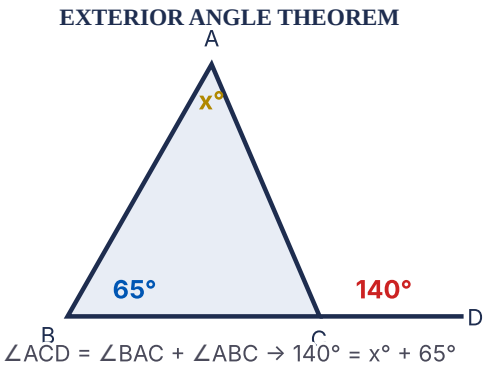
Answer: _____

3 Triangle Theorems

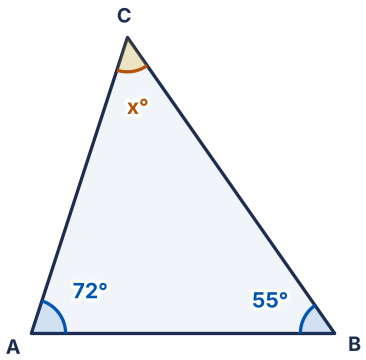
🔑 **TRIANGLE ANGLE SUM:** The three interior angles of a triangle always add up to 180°.

🔑 **EXTERIOR ANGLE THEOREM:** An exterior angle equals the two far inside angles added together. These are the **remote interior angles**.



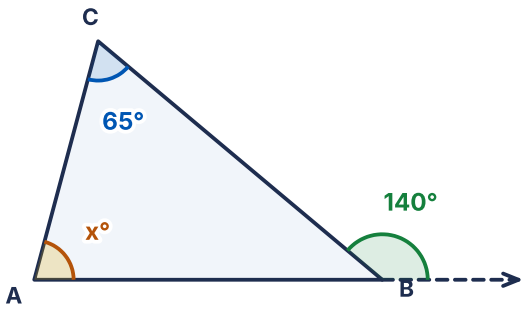


Problem 1 — Triangle Angle Sum: x + 72 + 55 = 180



Solve:

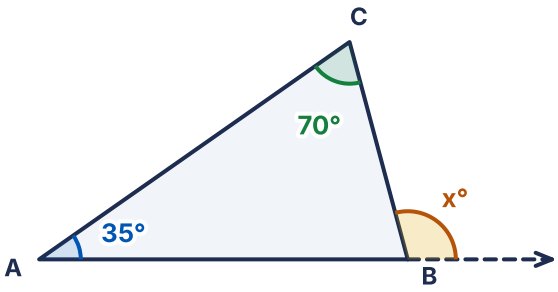
Problem 2 — Exterior Angle Theorem: x + 65 = 140



the outside angle = the two far inside angles

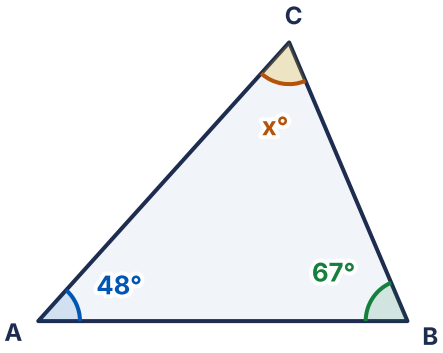
Solve:

ERROR ANALYSIS Jordan's Exterior Angle Mistake — A triangle has interior angles 35° and 70°. Exterior angle = x. Jordan wrote `x = 70 − 35 = 35°`.



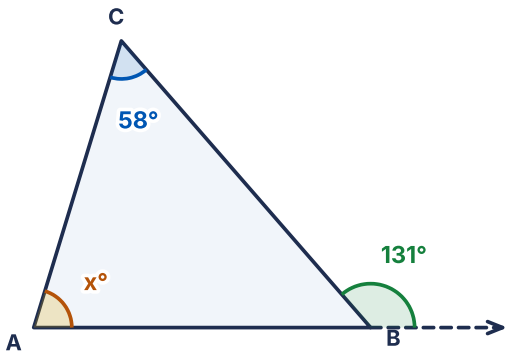
✏ Practice: Triangle Theorems

1. ★ Try it Find x.



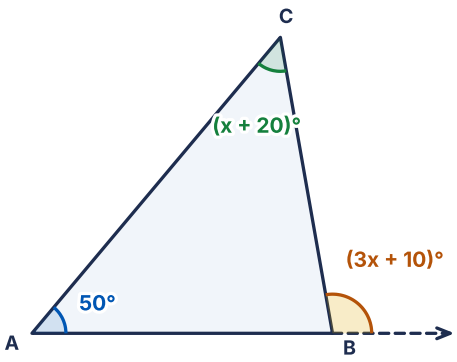
Answer: _____

2. ★★ On level The exterior (outside) angle is 131°. Find x.



Answer: _____

3. ★★★ Challenge Find x. Then find the measure of the exterior angle, (3x + 10)°.



Answer: _____

4 Angle-Angle (AA) Similarity

📖 **Words to know**

Similar: same shape, maybe a different size.

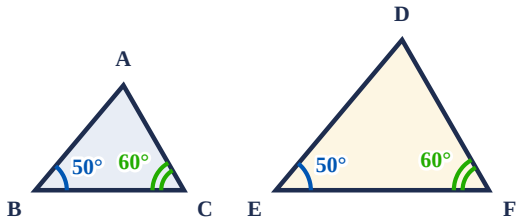
Scale factor: the number you multiply each side by to get the matching side.

In proportion: every side is multiplied by the same scale factor.

🔑 **SIMILAR TRIANGLES:** Two triangles are *similar* (written △ABC ~ △DEF) when they have the same shape but not necessarily the same size. Corresponding angles are equal and corresponding sides are in proportion.

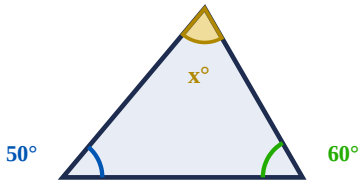
🔑 **ANGLE-ANGLE (AA) CRITERION:** If two angles of one triangle are equal to two angles of another triangle, then the triangles are similar. You only ever need to check TWO pairs of angles.

ANGLE-ANGLE (AA) CRITERION



matching marks = equal angles → △ABC ~ △DEF

WHY TWO ANGLES ARE ENOUGH

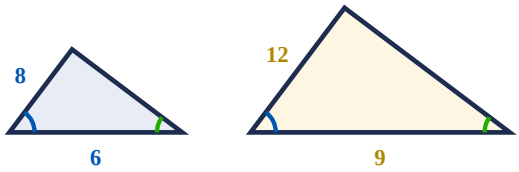


50° + 60° + x° = 180° → x = 70°

the third angle is forced, so all three match

🔑 **WHY TWO ANGLES ARE ENOUGH:** The angles of a triangle add to 180°. If two pairs match, the third pair must match too.

SIMILAR → SIDES IN PROPORTION

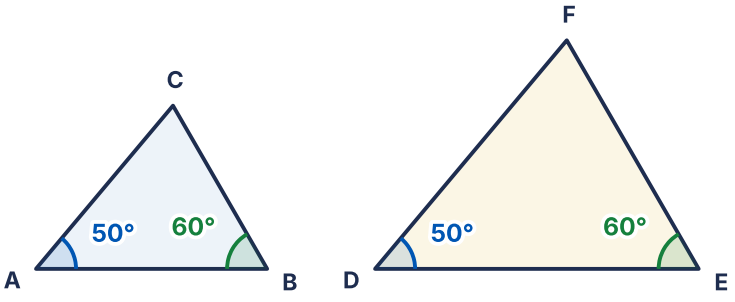


9 ÷ 6 = 1.5 and 12 ÷ 8 = 1.5

scale factor 1.5

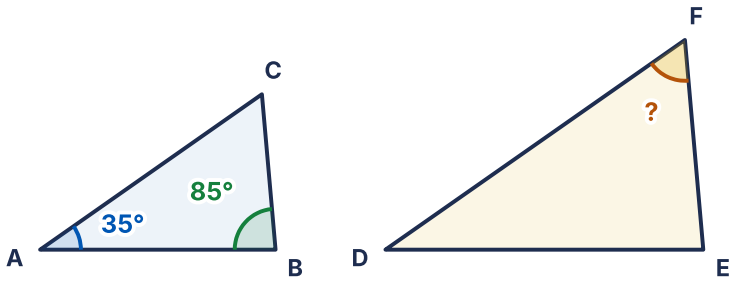
📌 **WATCH OUT:** AA needs two pairs of angles that *match each other* — not simply two angles given in each triangle.

Problem 1 — Are they similar? ∠A = 50°, ∠B = 60° and ∠D = 50°, ∠E = 60°



Decide:

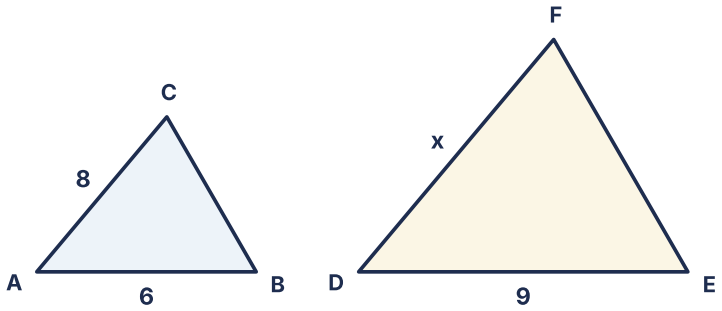
Problem 2 — △ABC ~ △DEF with ∠A = 35° and ∠B = 85°. Find ∠F.



F matches C

Solve:

Problem 3 — Similar triangles: sides 6 and 8 match sides 9 and x. Find x.



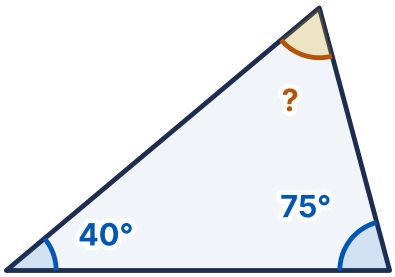
6 → 9 and 8 → x

Solve:

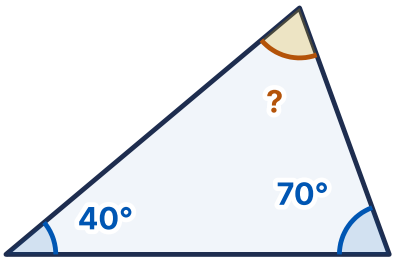
ERROR ANALYSIS Riley's AA Mistake — Triangle 1 has angles 40° and 75°. Triangle 2 has angles 40° and 70°. Riley wrote

> "Each triangle shows two angles, so they are similar by AA."

Triangle 1

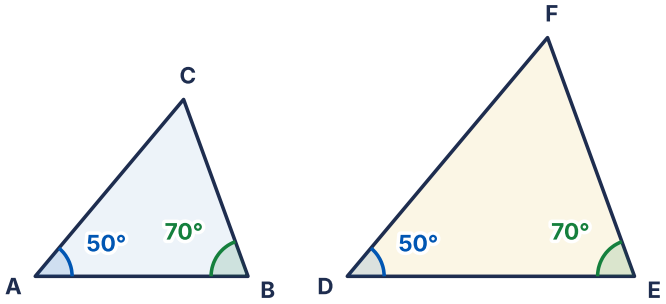


Triangle 2



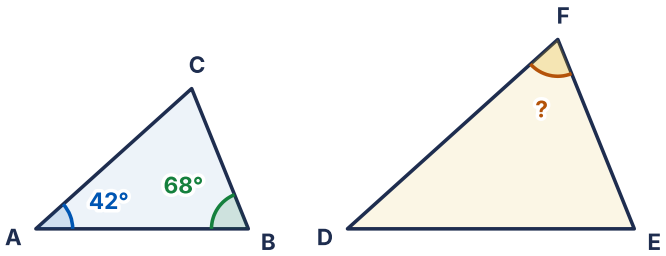
✏ Practice: AA Similarity

1. ★ Try it Are these two triangles similar?



- ○ Yes — two pairs of angles match
- ○ No — they are different sizes
- ○ You can't tell

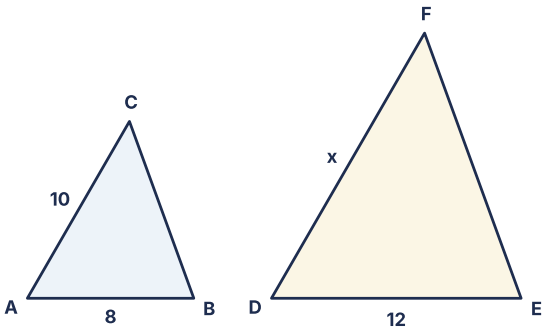
2. ★★ On level △ABC ~ △DEF. Find the measure of ∠F.



F matches C

Answer: _____

3. ★★ On level The triangles are similar. Side 8 matches side 12, and side 10 matches side x. Find x.

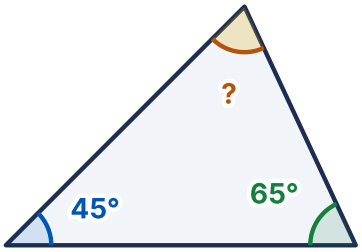


8 → 12 and 10 → x

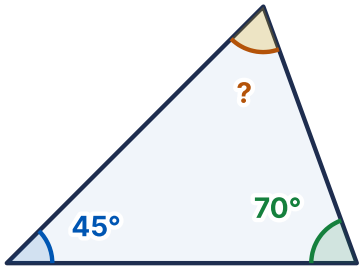
Answer: _____

4. ★★★ Challenge Triangle 1 shows 45° and 65°. Triangle 2 shows 45° and 70°. Sam says they are NOT similar. Is Sam right?

Triangle 1



Triangle 2



- ○ Yes — they are not similar
- ○ No — they are similar

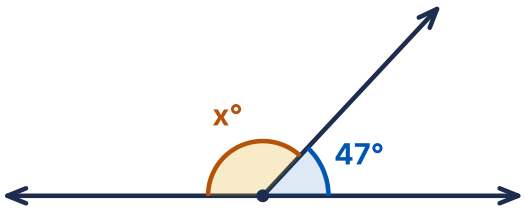
✏ Part 2 — Practice, Score 3

📌 **Show every step on its own line.** Box your final answer.

Every problem has a level: Score 3 Score 4 Score 5. Start with Score 3, then move up.

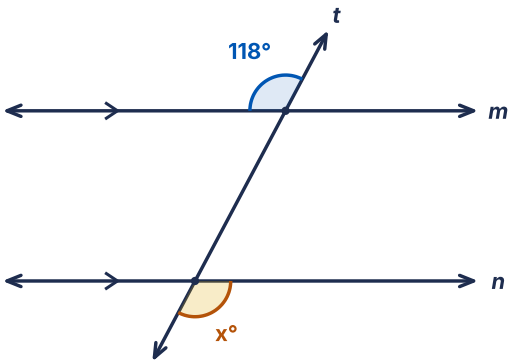
Score 3 Start here

1. Score 3 The two angles are supplementary. Find x.



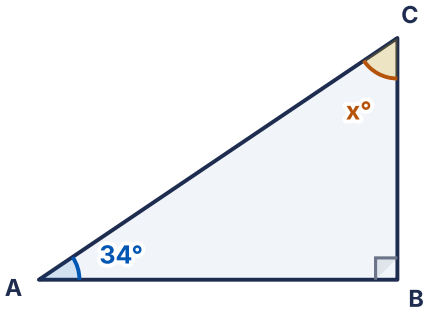
Answer: _____

2. Score 3 Lines m and n are parallel. Find x.



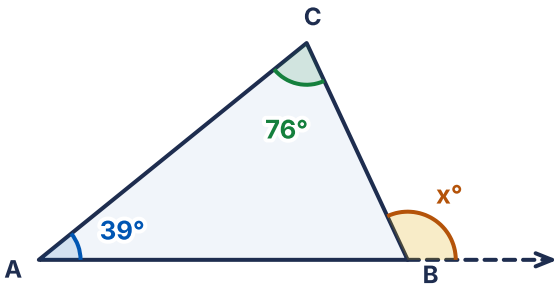
Answer: _____

3. Score 3 Find x. (The square corner is a right angle.)



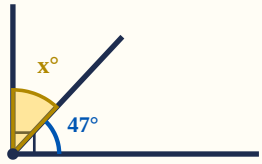
Answer: _____

4. Score 3 Find the exterior angle x.



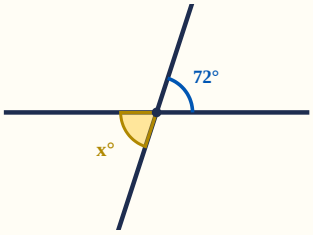
Answer: _____

Score 3 5) Complementary Angles, given 47°. x = ?



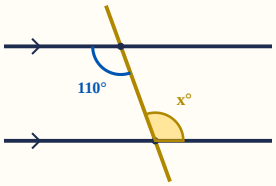
x = _____

Score 3 6) Vertical Angles, given 72°. x = ?



x = _____

Score 3 7) Alternate Interior, given 110°. x = ?

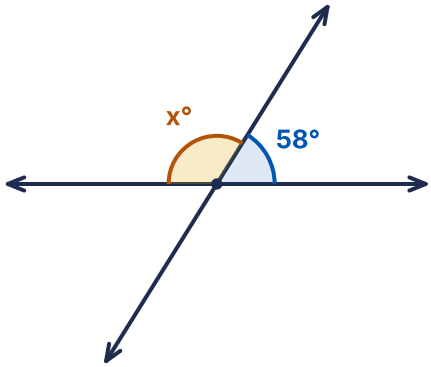


x = _____

Part 2 — Practice, Score 4

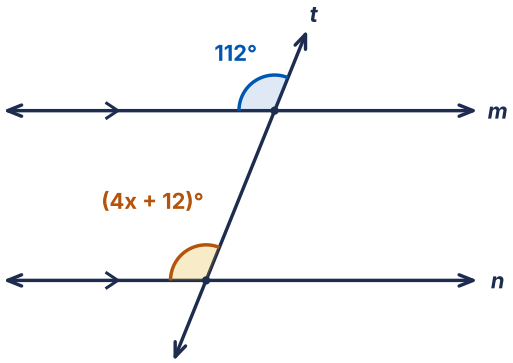
Score 4 **On grade level**

1. Score 4 Two lines cross. Find x.



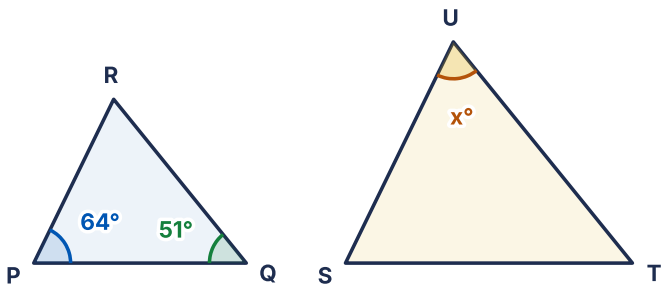
Answer: _____

2. Score 4 Lines m and n are parallel. Find x.



Answer: _____

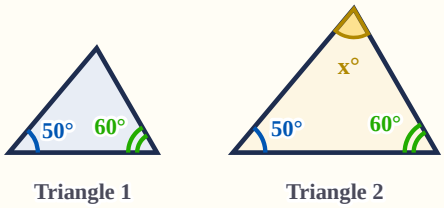
3. Score 4 △PQR ~ △STU. Find x.



U matches R

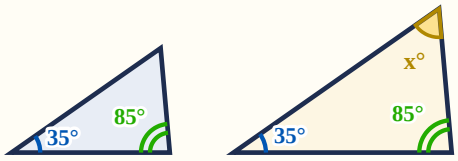
Answer: _____

Score 4 4) AA Similarity — two angles match: 50° and 60°. x = ?



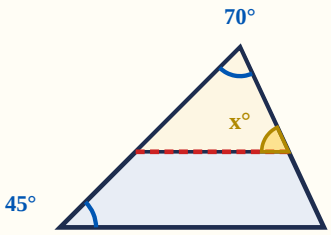
Triangle 1 Triangle 2

Score 4 5) △ABC ~ △DEF, given 35° and 85°. x = ?



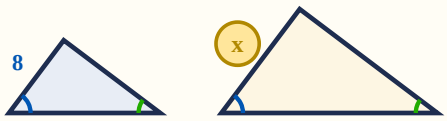
Triangle 1 Triangle 2

Score 4 6) Parallel line inside a triangle, given 70° and 45°. x = ?



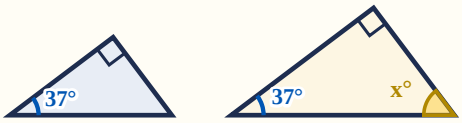
the dashed line is parallel to the base

Score 4 7) Similar triangles — sides 6 and 8 match 9 and x. x = ?



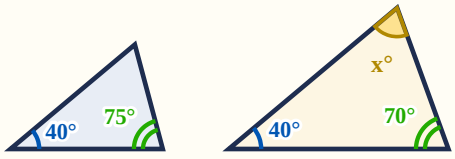
6 Triangle 1 9 Triangle 2

Score 4 8) Similar right triangles, given 37°. x = ?



Triangle 1 Triangle 2

Score 4 9) Triangle 1: 40°, 75°. Triangle 2: 40°, 70°. Find x, Triangle 2's third angle.

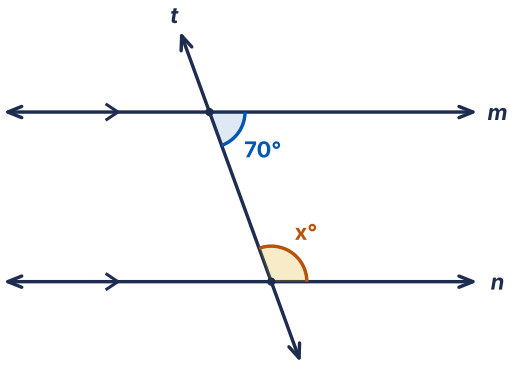


Triangle 1 Triangle 2

Part 2 — Practice, Score 5

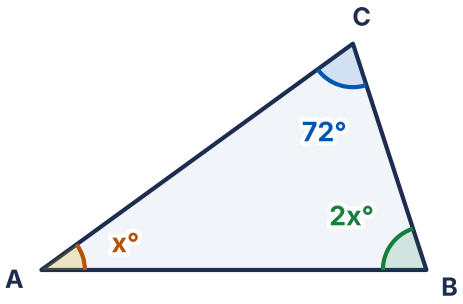
Score 5 Challenge

1. Score 5 Lines m and n are parallel. Maya says x = 70 because the angles "look the same." Find the correct value of x.



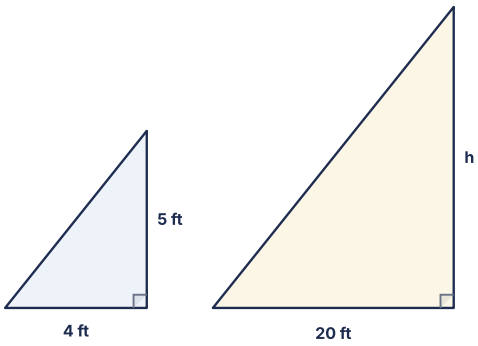
Answer: _____

2. Score 5 Find x.



Answer: _____

3. Score 5 A 5 ft student casts a 4 ft shadow. At the same time, a tree casts a 20 ft shadow. The two triangles are similar (AA). How tall is the tree?



student and shadow → tree and shadow (not to scale)

Answer: _____